INTERNSHIP REPORT

ON

DARKTRACE X CHATBOT

Submitted To

School of Computer Science and Engineering
IILM University, Greater Noida, U.P.

In partial fulfilment of the requirement of degree of

B.Tech CSE

By

Roll Number of Student: CS-2341762

Name of Student: PARUL BHATIA

Batch: 2026–27

2026–27

CANDIDATE'S DECLARATION

I, PARUL BHATIA, do hereby declare that the internship report titled "DarkTrace X Chatbot" has been prepared by me to fulfil the requirement for the award of the degree of Bachelor of Technology in Computer Science and Engineering. The report documents a conceptual and academic project proposal developed in the context of my internship and does not claim implementation, deployment, datasets, performance metrics, or integrations that are not supported by the supplied materials. All external sources used in preparing the report have been acknowledged. I affirm that this report is my own work and has not been submitted to any other institute for any degree or diploma requirement. I shall be solely responsible for any copyright violation in this regard.

Signature

PARUL BHATIA

Roll No. CS-2341762

ACKNOWLEDGEMENT

I express my sincere gratitude to Diligent Global Tech Consulting Pvt. Ltd. for providing me with the opportunity to undertake an internship as an AI Intern in the Technology and AI Department. The internship context enabled me to connect classroom learning in computer science with the structured analysis of an AI-enabled cybersecurity-awareness concept.

I am thankful to the organisation and its mentors for their guidance, constructive feedback, and encouragement during the internship period from 8 June 2026 to 31 August 2026. I also thank IILM University, the School of Computer Science and Engineering, and the faculty members for their academic support and for encouraging disciplined technical documentation.

Finally, I thank my parents and all those who supported me throughout the internship and the preparation of this report. Their encouragement helped me complete this work with sincerity, care, and professional responsibility.

Parul Bhatia

Signature

PARUL BHATIA

Roll No. CS-2341762

INTERNSHIP COMPLETION CERTIFICATE

The following certificate is reproduced as supplied. Its factual information has not been edited. It confirms the internship role, organisation, period, department, location, and performance rating.

DILIGENT GLOBAL TECH CONSULTING PVT. LTD.

Registered office: North Wing, Vth Floor, JVP Building, Plot No. S, Software Unit Layout, Hitech City, Madhapur, Hyderabad, Telangana - 500081, India.

CERTIFICATE

OF INTERNSHIP COMPLETION

This is to certify that

Ms. Parul Bhatia

Has successfully completed an Internship as an **AI INTERN** at **Diligent Global Tech Consulting Pvt. Ltd.**

The internship program duration was from: **8th June 2026 to 31st August 2026.**

During her tenure, she demonstrated exceptional skills in **Artificial Intelligence**, including **Machine Learning** model development and data analysis, with a focus on commitment and diligence, as highlighted in her provisional offer. She consistently exhibited a professional demeanor, curiosity, and strong analytical capabilities. All training requirements, assignments, and tests as defined by the company and detailed in her offer letter have been cleared and successfully met.

Details of Internship:

- **Role**: AI Intern
- **Department**: Technology and AI
- **Start Date:** 8th June 2026
- **End Date:** 31st August 2026
- **Location:** Diligent Global Tech Consulting, Hitech City, Hyderabad, India

Her performance during the training program was rated as **OUTSTANDING.**

In Witness Whereof, this certificate is awarded.

Dated: **31st August 2026**

Huma

Authorised Signatory
(Signature of Huma)
Huma
HR Department
For Diligent Global Tech Consulting Pvt. Ltd.

SEAL

Diligent Global
Consulting with Commitment

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5. PROJECT DESCRIPTION

5.1 Introduction

Cybersecurity awareness depends not only on technical controls but also on whether users can recognise risk and act appropriately. Phishing and related social-engineering attempts may arrive through email, text messages, phone calls, social media, or other channels. NIST guidance recommends teaching users how to identify and report phishing, avoiding interaction with suspicious messages and links, enabling multi-factor authentication, changing affected passwords, and following incident-response procedures [1]. These practices motivate an assistant that explains suspicious digital activity in a user-oriented manner.

DarkTrace X Chatbot is a conceptual AI-powered conversational assistant for cybersecurity and digital-threat awareness. Its proposed purpose is to let a user describe a suspicious message, link, login request, device behaviour, or online interaction in natural language. The system would then organise the description, identify relevant indicators, classify the broad awareness category, and provide a cautious explanation with safe next steps. The assistant is designed as an awareness and triage aid, not as a replacement for a security operations team, incident responder, or forensic investigation.

5.1.1 Conceptual User Need

The user need can be expressed as a translation problem: technical security knowledge must be converted into a response that is timely, comprehensible, and actionable without creating false certainty. A useful response should distinguish what the user observed from what can reasonably be inferred. It should also request only the minimum information needed for guidance and warn users not to paste passwords, private keys, one-time codes, or other secrets into a chat interface.

5.1.2 Proposed Conversation Model

Conversation Stage	Purpose	Illustrative Output
Describe	Capture the observation in free-form language.	“I received a password-reset message I did not request.”
Clarify	Ask a focused question when context is insufficient.	“Did the message contain a link or attachment?”

Interpret	Map the description to awareness categories and indicators.	Possible impersonation or credential-phishing indicators.
Guide	Explain safe next steps with uncertainty language.	Avoid the link; verify through an official channel.
Escalate	Recommend human or organisational support for serious cases.	Contact the security team or service provider.

5.1.3 Design Principles

• Clarity: responses should use plain language and explain security terms when they are necessary.

• Caution: the system should communicate uncertainty rather than claim that an event is definitely safe or malicious.

• Privacy: the user should be advised not to disclose credentials, secrets, or unnecessary personal data.

• Traceability: future implementations should record the response basis and knowledge source used.

• Human oversight: serious incidents, suspected compromise, and high-impact decisions should be escalated to qualified personnel.

5.2 Organization Profile

Diligent Global Tech Consulting Pvt. Ltd. is the organisation named in the supplied internship completion certificate. The certificate identifies the registered office as North Wing, VII Floor, JVP Building, Plot No. S, Software Unit Layout, Hitech City, Madhapur, Hyderabad, Telangana – 500081, India. It records the student's role as AI Intern in the Technology and AI Department and identifies the internship location as Diligent Global Tech Consulting, Hitech City, Hyderabad, India.

The certificate states that the internship programme ran from 8 June 2026 to 31 August 2026 and that the student's performance was rated OUTSTANDING. Since no additional corporate profile material was supplied, this report does not add unsupported claims about the company's history, founders, clients, revenue, workforce, products, or achievements.

Organisation Diligent Global Tech Consulting Pvt.

Ltd. Role AI Intern Department Technology and AI Start Date 8 June 2026 End Date 31 August 2026 Location Hitech City, Hyderabad, India Performance OUTSTANDING

5.3 Problem Statement

Users frequently face suspicious digital situations before they understand the relevant security concepts. A message may imitate a trusted organisation, create urgency, request credentials, or direct the user to an unfamiliar website. The user must make a decision quickly, yet technical guidance is often dispersed across policy pages, advisories, or specialist tools. This creates a usability and awareness gap between the availability of security knowledge and the user’s ability to apply it in context.

The problem addressed by the proposed DarkTrace X Chatbot is the design of an accessible conversational interface that can interpret user descriptions of suspicious digital activity and return structured, responsible guidance. The system should help the user understand possible indicators and decide on safer next actions while avoiding unsupported conclusions. The problem is therefore not defined as fully automated threat detection; it is defined as contextual awareness assistance.

5.3.1 Risk Categories Addressed

Category Common Indicators Awareness Response Direction Unexpected request, Avoid the link, verify Phishing urgency, suspicious link, independently, report the impersonated sender. message.

Request for password, Do not disclose secrets; Credential theft MFA code, or account change affected recovery details. credentials if exposed. Unexpected attachment, Do not open; scan the Malware delivery download prompt, or device and contact executable file. support if opened.

Brand or person identity Confirm through an Impersonation appears inconsistent or official channel before unusually urgent. acting.

Category Common Indicators Awareness Response Direction Unrecognised login, Secure the account, Account compromise password reset, or review activity, and security notification. escalate if necessary.

5.4 Project Objectives

The objectives define the proposed system’s academic and technical direction. They are written as design objectives rather than evidence of completed production implementation.

• Design a conversational interface for cybersecurity and digital-threat awareness.

• Model natural-language input processing through intent recognition, entity

extraction, normalisation, and context handling.

• Organise threat-awareness guidance into understandable categories such as

phishing, impersonation, credential theft, malware delivery, and account compromise.

• Generate structured responses that separate observed facts, possible indicators,

confidence limits, recommended safe actions, and escalation paths.

• Define an architecture that can later connect to validated knowledge sources under

controlled security and privacy requirements.

• Establish conceptual evaluation criteria for clarity, relevance, safety, consistency,

privacy, and appropriate uncertainty.

Objective ID Objective Indicative Evaluation Question Can a non-specialist understand the response O1 Accessible conversation without specialist vocabulary? Does the system use the O2 Context handling user’s description without losing relevant details?

Does the response identify O3 Threat awareness plausible indicators without overclaiming? Does the response provide O4 Safe action practical, low-risk next steps? Does the system protect O5 Responsible AI privacy and support human escalation?

5.5 Scope of the Project

The scope includes the conceptual requirements, user interaction model, information-processing pipeline, logical architecture, methodology, proposed technology stack, safety controls, and evaluation plan for DarkTrace X Chatbot. It covers awareness-oriented questions about suspicious messages, links, login requests, account notifications, and general digital-safety concerns.

The scope excludes claims of a deployed product, live users, measured accuracy, real-time threat detection, a production database, a specific cybersecurity API, a trained model, or an operational integration. The report also excludes autonomous blocking, forensic attribution, legal conclusions, and guaranteed identification of malicious content. These exclusions are essential because the supplied materials describe the internship and project concept but do not provide implementation evidence.

In Scope Out of Scope Conceptual conversation flow and user Production deployment or public release prompts Claiming a specific trained model or NLP and AI processing design accuracy Threat-awareness categories Forensic investigation or attribution Real users, datasets, or performance Logical architecture and data flow metrics Safety, privacy, and escalation Autonomous incident response or requirements account actions

5.6 Technologies and Tools Used

Conversational AI systems combine natural language processing, machine learning, and foundation-model capabilities to interpret and generate human-language interaction [4]. For DarkTrace X Chatbot, these technologies are proposed as a modular stack. The table below distinguishes the conceptual role of each component from confirmed internship facts.

Component Technical Role Design Status Considerations Application Modular functions, orchestration, text input validation, Conceptual / Python processing, logging, and proposed validation, and testability.

experimentation. NLP pipeline Normalisation, Ambiguity Conceptual / language handling, proposed

Component Technical Role Design Status Considerations detection, intent classification, multilingual entity extraction, extension, and and context fallback responses. handling. Drafting Prompt controls, explanations and Transformer / grounding, output Conceptual / identifying GenAI layer validation, and proposed relationships in refusal behaviour.

user descriptions. Dialogue state, routing, State isolation, Chatbot clarification Conceptual / timeout handling, framework questions, and proposed and safe defaults. session management.

Connect interface, processing Authentication, services, and rate limiting, Conceptual / Backend / API approved encryption, and proposed knowledge least privilege. sources. Store reviewed awareness Versioning, source guidance, attribution, review Conceptual / Knowledge layer categories, ownership, and proposed examples, and update process.

response templates. Provide chat input, Accessibility, response sections, responsive layout, Conceptual / Web interface warnings, and and avoidance of proposed escalation secret collection.

prompts.

5.6.1 NLP Processing Pipeline

A proposed NLP pipeline begins with input validation and normalisation. The system may remove unnecessary formatting, detect the language, and identify whether the message contains a question, an incident description, or a request for action. Intent classification can then map the query to a broad task such as “assess suspicious link,” “explain account alert,” or “report possible phishing.” Entity extraction may identify a sender, domain, message channel, account type, or attachment description. Context handling can retain relevant information across follow-up questions while applying privacy limits.

The pipeline should not treat linguistic confidence as proof of maliciousness. A classifier may be uncertain because a user provides incomplete information, uses unfamiliar terminology, or describes a novel attack. Therefore, the response layer should combine model output with deterministic safety rules, approved knowledge content, and an uncertainty-aware template.

5.6.2 Generative AI and Retrieval Considerations

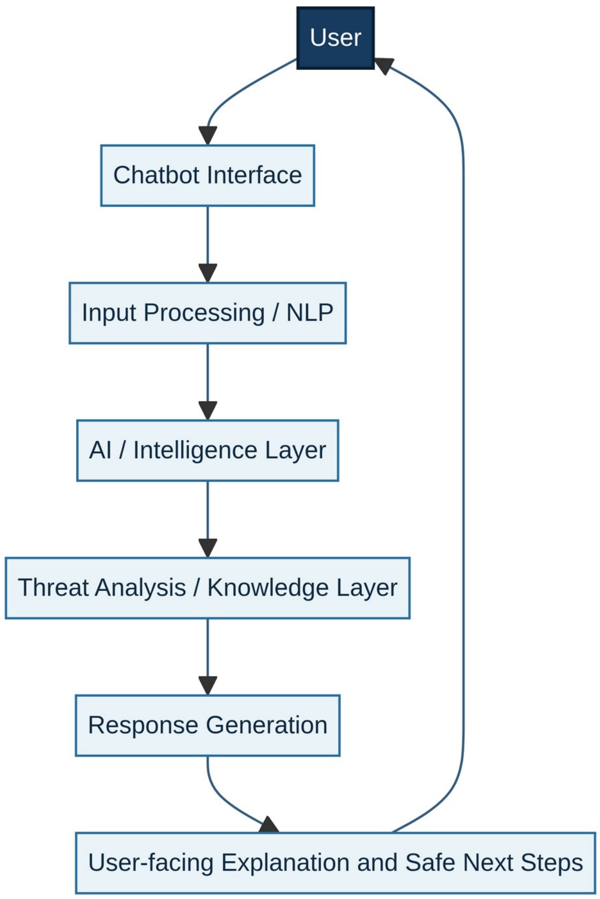
A future implementation could use a retrieval-augmented design in which approved awareness documents are retrieved before response generation. Retrieval may improve source grounding, but it introduces additional requirements: documents must be reviewed and versioned; retrieved text must be treated as data rather than instructions; and prompt injection or malicious content in retrieved material must be considered. OWASP identifies prompt injection, insecure output handling, sensitive information disclosure, excessive agency, and supply-chain concerns among the risks relevant to LLM applications [3].

For this reason, the proposed system should validate generated output before displaying it. A validator could check that the response does not request passwords or one-time codes, does not provide unsafe operational instructions, includes an uncertainty statement when evidence is incomplete, and routes high-risk situations to human support. These are design requirements, not claims that the controls were implemented during the internship.

5.7 System Architecture

The proposed architecture separates user interaction, language processing, intelligence, threat-awareness knowledge, and response generation. Separation improves maintainability and makes it possible to apply different controls at each stage. The interface should collect the minimum information required. The processing layer should transform text into structured signals. The intelligence layer should combine those signals with approved guidance. The response layer should apply safety checks before returning the result.

Figure 5.1: Proposed logical architecture of DarkTrace X Chatbot



5.7.1 Architectural Layers

Layer Input Output Primary Control User description Privacy notice and Validated text User / Interface and selected secret-data request. context. warning. Intent, entities, Input validation Input Processing / Normalised channel, and and uncertainty NLP request.

context. handling. Structured signals Candidate Prompt constraints AI / Intelligence and conversation interpretation and and least privilege. state. response plan. Relevant Threat Analysis / Candidate awareness Approved, Knowledge interpretation. guidance and versioned sources.

indicators. Draft explanation Response Guidance and Output validation and safe next Generation response plan. and refusal rules. steps. Explanation, User-facing Validated Human oversight actions, and Response response. for serious cases.

escalation.

5.7.2 Data Flow and Privacy Boundaries

The preferred data flow is minimal and purpose-limited: the user submits a description, the system processes the text, the knowledge layer supplies relevant guidance, and the response is returned. A future implementation should define retention rules before storing conversation history. Sensitive data should be masked or rejected where possible. Logs should record operational events without retaining unnecessary message content. Access to any stored information should be authenticated, authorised, and auditable.

5.8 Methodology

The methodology follows a staged academic workflow. It is designed to move from problem understanding to conceptual design and evaluation without implying that a production system was built.

Figure 5.2: Proposed nine-step project methodology

5.8.1 Methodology Stages

Stage Activities Expected Artefact Identify users, use cases, Requirement 1. Requirement Analysis safety expectations, and specification. privacy boundaries.

Describe the awareness Problem definition and 2. Problem Identification gap and prioritise threat category list.

categories. Define modules, data 3. System Design flow, interfaces, and trust Logical architecture. boundaries. Specify intents, entities, 4. Conversational / NLP dialogue states, and Conversation schema.

Design clarification questions. Define reasoning prompts, Response policy and 5. AI Intelligence Design response templates, and prompt design. uncertainty rules. Map categories to

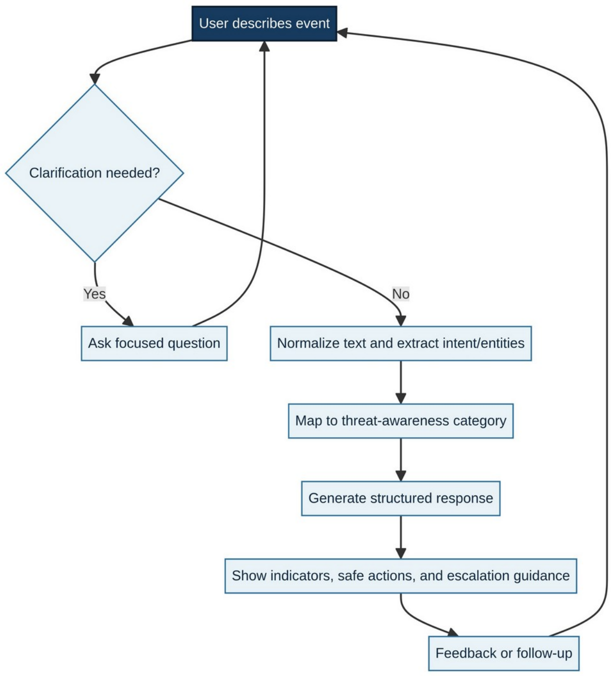
6. Threat-Awareness

indicators, actions, and Knowledge model. Logic escalation guidance. Design chat layout, Interface wireframe or 7. Interface Design warnings, response prototype specification. sections, and accessibility.

Review scenarios for 8. Conceptual Evaluation clarity, relevance, safety, Evaluation matrix. privacy, and consistency. Record decisions, limitations, references, 9. Documentation Final internship report.

and future implementation needs.

Figure 5.3: Proposed chatbot interaction and clarification flow



5.8.2 Conceptual Evaluation Framework

Because no implementation or test results were supplied, evaluation is defined as a future plan rather than reported evidence. A scenario-based review could present representative user descriptions and assess whether the system identifies the intended category, avoids unsafe certainty, provides relevant action, protects sensitive information, and recommends escalation when appropriate.

Evaluation Dimension Review Question Evidence to Collect in Future Can users understand the User comprehension Clarity response? feedback.

Evaluation Dimension Review Question Evidence to Collect in Future Does the response address Scenario-to-response Relevance the described situation? review. Does the response avoid Security review and red- Safety harmful or overconfident team cases.

guidance? Does the interaction Data-flow and retention Privacy minimise sensitive data audit. collection? Do similar scenarios Repeated scenario Consistency receive comparable comparison.

guidance? Are serious or uncertain Human-review decision Escalation cases routed log. appropriately?

5.8.3 Security and Responsible-AI Controls

NIST describes the AI Risk Management Framework as a voluntary approach for incorporating trustworthiness into the design, development, use, and evaluation of AI systems. It highlights transparency, reliability, privacy, and human oversight [2].

These principles translate into concrete requirements for the proposed chatbot: explain the basis of guidance where feasible; communicate uncertainty; avoid retaining unnecessary personal data; provide a path to human assistance; and review model, prompt, and knowledge-source changes.

Risk Possible Failure Proposed Control User or retrieved content Instruction hierarchy, Prompt injection attempts to override content separation, and system rules. output validation.

Generated advice Allow-listed response Insecure output encourages unsafe patterns and human actions. review for high-risk cases. Warnings, input filtering, User enters credentials or Sensitive disclosure masking, and minimal private information.

retention. System invents a fact, Grounded knowledge, Hallucinated guidance source, or technical citations, uncertainty conclusion. language, and refusal. System takes or Read-only awareness role Excessive agency recommends irreversible and explicit human action. confirmation.

Versioned sources, Guidance becomes Knowledge drift periodic review, and outdated or inconsistent. change logs.

5.9 Expected Outcomes

The expected outcomes are stated as potential benefits of a properly implemented and validated system. They are not measured outcomes from the internship.

• Improved user interaction through a natural-language interface that accepts

questions in ordinary language.

• Faster understanding of cybersecurity-related queries by presenting relevant

indicators and definitions together.

• Conversational assistance through clarification questions and context-aware

follow-up responses.

• Better awareness of suspicious digital activity such as phishing, impersonation,

credential requests, and unsafe attachments.

• Structured and understandable responses that separate observations, possible risk

indicators, safe actions, and escalation guidance.

• A foundation for future integration with validated cybersecurity knowledge

sources and carefully governed tools.

5.9.1 Limitations and Assumptions

The central limitation is that this report documents a conceptual project rather than a verified deployment. The quality of a future system would depend on the coverage and maintenance of its knowledge sources, the behaviour of the chosen model, the quality of evaluation scenarios, and the controls surrounding privacy and access. The chatbot could support awareness, but it could not guarantee that a message is harmless or malicious from incomplete user-provided information.

5.9.2 Future Scope

• Add reviewed and versioned cybersecurity-awareness content with source attribution.

• Develop controlled prototypes for intent classification, entity extraction, and response validation.

• Evaluate the system with representative scenarios and human reviewers.

• Add privacy-preserving logging, audit trails, and role-based access control.

• Study multilingual support and accessibility for users with different language needs.

• Explore integrations with organisational reporting or ticketing workflows only after security review and explicit human approval.

5.10 Certificates of Completion and Communication Mail Proof

The original internship completion certificate is included in Section 4 of this report. It states that Ms. Parul Bhatia successfully completed an internship as an AI Intern at Diligent Global Tech Consulting Pvt. Ltd. from 8 June 2026 to 31 August 2026. It identifies the Technology and AI Department, the Hyderabad location, and an OUTSTANDING performance rating.

No communication-mail proof was supplied with the source materials. Accordingly, this report does not fabricate or represent any email as attached. If the university requires email evidence, the student may insert an authentic communication record in this section after removing unrelated personal information and preserving the original date, sender, recipient, and subject details.

Embedded Figures Preserved from Source PDF

The following visual assets were extracted from the supplied PDF and retained for reference. Their captions should be updated if the source report identifies them more specifically.

Table No.	Control	Purpose
Table 5.7	Privacy filter	Discourages disclosure of passwords, keys, and one-time codes.

Table 5.8	Uncertainty language	Separates observations from possible interpretations.
Table 5.9	Human escalation	Routes suspected compromise or high-impact cases to qualified support.

Additional Technical Figures

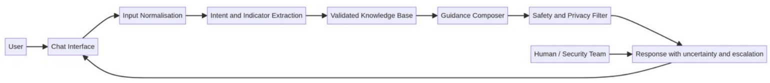


Figure 5.4 Additional DarkTrace X logical architecture

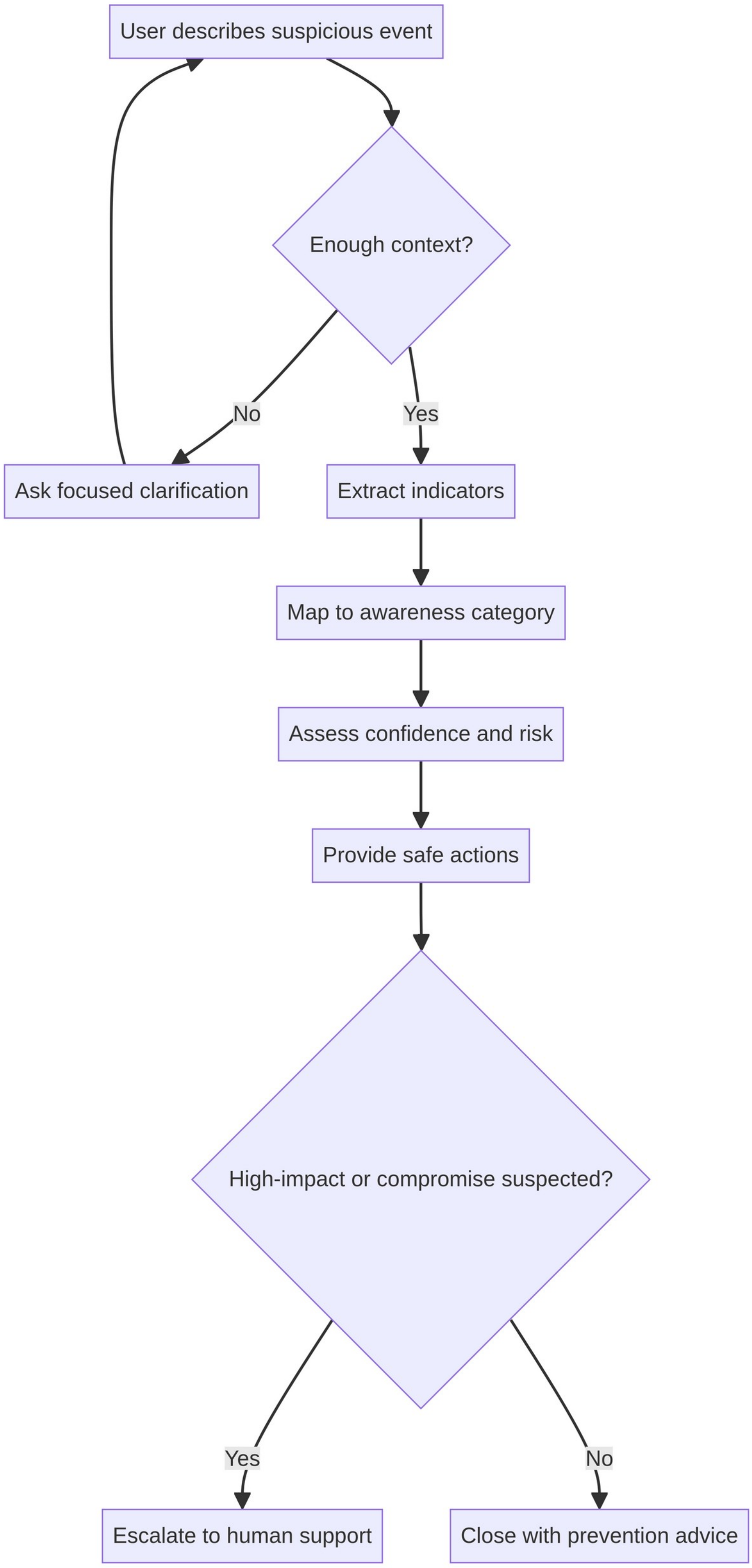


Figure 5.5 Additional conversation and escalation workflow

Figure 5.6 Additional safety, privacy, and review workflow

CHAPTER 6 — BIBLIOGRAPHY / REFERENCES

[1] National Institute of Standards and Technology. “Phishing Guidance and Awareness Resources.” https://www.nist.gov/

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[3] Python Software Foundation. “Python 3 Documentation.” https://docs.python.org/3/

[4] Streamlit. “Streamlit Documentation.” https://docs.streamlit.io/